

Developmental trajectories of school burnout: Evidence from two longitudinal studies

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ARTICLE INFO

Article history:

Received 16 November 2013

Received in revised form 15 August 2014

Accepted 27 October 2014

Available online xxxx

Keywords:

Adolescence

School burnout

Growth mixture modeling

School transitions

Demands–resources model

ABSTRACT

To examine school burnout trajectories, in Study 1, 15-year-old adolescents ($N = 614$) completed the School Burnout Inventory twice during their final term of comprehensive school and twice after the transition to upper secondary high school or vocational school. In Study 2, school burnout in 17-year-old adolescents was measured twice annually ($N = 575$) during their upper secondary education. In Study 1, growth mixture modeling for school burnout yielded a four-latent-group solution: 60% of the adolescents showed a low and stable level of school burnout, 29% increasing burnout, 3% strongly increasing burnout and 8% high-decreasing school burnout. In Study 2, we identified two latent groups of school burnout: 94% moderate and slightly decreasing, and 6% high-increasing. In support of the demands–resources model during the educational transition, more changes were observed in the trajectories of school burnout, while during the stable post-transitional period, the trajectories of school burnout were mostly stable.

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A few studies have adopted a person-oriented approach and examined the heterogeneous trajectories of school burnout during the educational transitions that occur in middle and late adolescence (Jung & Wickrama, 2008). By drawing on data on two longitudinal studies, the first during the education transition period from comprehensive school to upper secondary schooling, and the second during the more stable educational stage of upper secondary schooling, the aim of the present study was to examine heterogeneity in the trajectories of school burnout during both transition and stable educational stages. Based on the demands–resources model (Demerouti, Bakker, Nachreiner, & Schaufeli, 2001; Salmela-Aro & Upadyaya, 2014), we predicted that during the educational transition period, we would identify heterogeneous trajectories showing changes – both increasing and decreasing – in school burnout, in accordance with increasing and decreasing demands, while during more stable educational stages, the school burnout trajectories would mainly differ only in their level of stability.

While the majority of young people manage adolescence without experiencing any severe problems, some have difficulties in adapting to the transitions and changes of this age period: the rates of negative and stressful life events rise dramatically during adolescent years (Hankin & Abrahamson, 2001). School burnout is a new phenomenon consisting of exhaustion due to school demands, a cynical and detached attitude toward one's school, and the feelings of inadequacy as a student (Salmela-Aro, Kiuru, Leskinen, & Nurmi, 2009a). School burnout can be

considered as a continuous phenomenon running from school-related stress to major burnout. In line with this, the present study focuses on school burnout symptoms as a continuum, and examines school burnout as a single concept consisting of the three theoretically-derived dimensions of school burnout.

Some research has been conducted on the course of school burnout throughout adolescence (Salmela-Aro, Savolainen, & Holopainen, 2009b): Overall, considerable stability has been found in the overall mean level of school burnout. However, during educational transitions, heterogeneous trajectories might appear. A key educational transition of adolescence in many European educational systems and a challenge for school adjustment is the transition to post-comprehensive education. This determines the quality and types of school-related demands, resources and learning opportunities among students and their exposure to different peers and teachers. We argue that educational transitions contribute to school burnout as educational transitions can be disruptive and demanding (Demerouti et al., 2001; Eccles & Midgley, 1989) and lead to heterogeneous pathways.

According to the demands–resources model (Demerouti et al., 2001; Salmela-Aro & Upadyaya, 2014), the fit between adolescents' needs and resources in the context of changing school-related demands influences their school-related burnout. Adolescents' emotional, cognitive, and social resources change as they mature (Eccles & Midgley, 1989). Based on the demands–resources model, optimal development takes place when there is a good fit between the demands and resources of the developing individuals and the opportunities afforded them by their social environments. Eccles (2004) argued that if schools are to provide the kind of social context that will continue to motivate students' engagement as they

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mature, then schools need to change in developmentally appropriate ways. To the extent that this does not occur, as they mature during adolescence, adolescents disengage from school first psychologically and then physically.

Recently, it has been shown that educational transitions may represent phases in which school burnout is susceptible to destabilization (Parker & Salmela-Aro, 2011). However, for some, transitions are opportunities for positive change, while for others these transitions can be quite disruptive. Nevertheless, moderate stability in school burnout was found across the transition to post-comprehensive education (Parker & Salmela-Aro, 2011). However, a major limitation of the previous studies is that they have all focused on examining the developmental course of school burnout at an average level within the population over time. For example, although latent growth modeling, which has been used in research on school burnout among adolescents (Salmela-Aro, Kiuru, & Nurmi, 2008), offers an efficient tool, it nevertheless assumes that the observations are all from one population. Thus, there is a need for the use of more person-oriented or idiographic methods, such as growth mixture modeling (Muthén & Muthén, 2000) to detect trajectories of school burnout over time.

Our aim was to identify distinct subgroups of adolescents who follow similar longitudinal trajectories of school burnout across their transition to post-comprehensive education, and during their upper secondary education. The hypotheses are presented in the context of the demands–resources model:

- (1) What kinds of developmental trajectories do adolescents show in school burnout during their transition to post-comprehensive education? We assumed that most would pass through this period with a low level of school burnout (Hypothesis 1a). This is based on earlier findings that a majority of young people manage adolescence without experiencing any severe problems, while some have difficulties in adapting to the transitions and to the changes of this life phase (Eccles, 2004). Based on the demands–resources model (Hypothesis 1b), we assumed that the students undergoing the transition would show heterogeneous pathways: some would experience an increase, and some a decrease in school burnout over the follow-up during the educational transition.
- (2) What kinds of developmental trajectories do adolescents show in school burnout during the stable stage of their upper secondary education? We assumed that most would pass through this period without burnout (Hypothesis 2a). Moreover, we assumed that the heterogeneous trajectories would differ mainly by level (Hypothesis 2b) and that, among a small minority, burnout would increase in line with the increasing demands encountered at the end of the upper secondary education (Hypothesis 2c).
- (3a) To what extent students' gender predict their school burnout trajectory membership?
- (3b) To what extent students' educational expectations predict their school burnout trajectory membership?
- (3c) To what extent students' latent trajectory membership in school burnout predict their fulfilled educational expectations at the end of the study? As this is the first study to examine this issue these analyses are exploratory and no hypotheses are stated.

1. Study 1

1.1. Method

Study 1 is part of the ongoing FinEdu study. At the beginning of the study, the participants were ninth-graders facing the transition to their post-comprehensive schooling. Students from eight schools in a medium-sized town (population = 88000) in Central Finland were recruited for the study ($N = 773$). Two measurements were carried out before their transition to either upper secondary education (academic track) or vocational education (vocational track): at the beginning

(Age = 15, Time 1) and at the end of the final term of the comprehensive school (Age = 16, Time 2). The next two measurements were carried out after their transition to their choice of post-comprehensive schooling (January 2005; Time 3, Age = 17 and January 2006, time 4, Age = 18), and the last measurement was carried out 2 years after that (Time 5, in 2008). At Time 1, 614 (289 females, 325 males) of the 773 students attending the schools participated in the study. During the next waves, as the students made the transition to either a vocational or upper secondary school track, the questionnaires were also given to possible new classmates, and thus the number of participants slightly rose. At Time 2, the number of participants was 630 (313 females, 317 males), at Time 3 it was 732 (359 girls, 373 boys), at Time 4, it was 619 (321 girls, 298 boys), and at Time 5, it was 589 (323 females, 266 males). The median age of the participants was 15 at the first measurement ($M = 15$; $SD = 0.34$).

In the attrition analyses, the adolescents who participated in the study at each measurement point were compared with those for whom data were missing at some measurement point. The results indicated no selection effect with respect to the school burnout variables. By using the missing data procedure (see details below in the description of the time strategy used in the analysis), we were able to supply data for all the participants ($N = 863$) in the analyses. The majority of the participants (99%) were Finnish-speaking.

1.2. Measures

The School Burnout Inventory (Salmela-Aro, Kiuru, Leskinen, & Nurmi, 2009a) consists of 9 items measuring three components of school burnout: (1) exhaustion at school, (2) cynicism toward the meaning of school and (3) sense of inadequacy at school to be rated on a 6-point scale (1 = *strongly disagree*; 6 = *strongly agree*). The total sum score was used. Cronbach's alpha reliability was Time 1 = .83; Time 2 = .86; Time 3 = .87, and Time 4 = .88.

Educational expectations (Time 1) were measured by asking the participants "If you think about your future, what is the highest degree you are going to attain?" coded as 1 = university degree, 2 = polytechnics, 3 = vocational school, and 4 = no degree/other.

Educational track after comprehensive school (Time 3) was measured by asking: (1) *Are you in education at the moment?* (1 = yes, 0 = no); (2) *If you are in education, what is the name of your school?* coded as: 1 = upper secondary high school (58.3%), 2 = vocational school combined with upper secondary high school courses (4.3%), 3 = vocational school (28.0%), 4 = tenth grade voluntary (6.9%), and 5 = outside formal education (2.5%). Educational track variable was created by comparing those on an academic track with those on a vocational track. Adolescents who were in vocational school and those in vocational school who were also doing upper secondary high school courses were considered vocational students.

Fulfilled educational expectations were measured by asking about current educational qualification and life situation at Time 5 (e.g., "What is your current employment study situation?"; "What is the highest degree you are going to attain?"). The answers were coded as follows: 1 = the highest expected degree is higher than the one completed or being studied for at the moment, and 2 = the highest expected degree is the same.

Gender 1 = girl, 2 = boy.

The first aim was to examine whether adolescents that show different developmental trajectories in school burnout during the transition to post-comprehensive education were examined by Growth Mixture Modeling (GMM; Muthén & Muthén, 2000). GMM provides an excellent tool for examining heterogeneity in growth by identifying homogeneous subgroups of participants that differ with respect to their developmental trajectories. It estimates mean growth curves for each identified class of participants and captures individual variation in these growth curves by estimation of the variance of the level and growth factor for each class (Muthén & Muthén, 2000).

Table 1

Study 1: sample correlation matrix, means, and variances for the whole sample ($N = 863$).

Variable	1.	2.	3.	4.
1. School burnout (T1)	1.00			
2. School burnout (T2)	0.53***	1.00		
3. School burnout (T3)	0.45***	0.47***	1.00	
4. School burnout (T4)	0.41***	0.34***	0.59***	1.00
Mean	2.48	2.52	2.49	2.63
Variance	0.78	1.24	0.86	0.87

*** $p < .001$.

The GMM analyses were carried out in two steps: (1) an overall latent growth model (LGM) was estimated for the whole sample (Muthén & Muthén, 1998–2012) and (2) GGM analyses were conducted (van Lier, Muthén, van der Sar, & Crijnen, 2004). LGM estimates different growth components, such as level (*intercept*), linear change (*linear trend*), and quadratic change (*quadratic trend*), from data consisting of at least three measurements. Besides estimating the means of these components the shape of change, LGM estimates also the variances of the level and growth components, and whether these variances are statistically significant. In the overall latent growth model, the loadings of the observed variables across Times 1 to 4 were fixed to 1 on the level factor and to 0, 1, 2 and 5 on the linear change factor. The residual variances were allowed to be freely estimated.

Second, GMM analyses were carried out. These analyses aimed to explore the extent to which distinct trajectory classes (groups of homogeneous subjects) could be identified on the basis of the growth factor means (i.e., means of level and linear change) and variances (i.e., variances of level and linear change) so that each class defines a different trajectory over time. The overall growth model was used as a basic model for searching for the optimal number of latent groups. A set of GMM analyses including different numbers of latent trajectory groups was carried out and their fit indices and class frequencies were compared. The solution that fitted the data best and was also a reasonable one in the light of their interpretation was chosen as the final mixture model.

Finally, in order to identify the possible antecedents and consequences of the burnout trajectory groups, the participants' current (Time 1) and fulfilled (Time 5) educational expectations, gender, and track were added separately to the final GMM model as covariates.

The analyses were performed by using the Mplus statistical package (Version 6, Muthén & Muthén, 1998–2010) with the missing data method, that is, the standard FIML approach (*full information maximum likelihood method*). This method uses all the available data in order to estimate the model, without imputing data. By using this data method, we were able to utilize all the available observations in the data set when estimating the parameters of the models. Since the variables

measuring school burnout were skewed, the parameters of the LGM and GGM models were estimated using the MLR estimator. The MLR produces robust standard errors and a chi-square test for missing data with non-normal outcomes by means of a sandwich estimator (Muthén & Muthén, 1998–2010). The goodness-of-fit of the estimated latent growth models was evaluated according to the following four indicators: (a) chi-square test, (b) Comparative Fit Index (*CFI*), (c) Root Mean Square Error of Approximation (*RMSEA*), and (d) Standardized Root Mean Square Residual (*SRMR*). We used the following indices to select the number of latent trajectory classes in the GMM: (a) the Bayesian information criterion (*BIC*), (b) *AIC*, (c) entropy values (range from 0 to 1, values close to 1 indicating a clear classification), (d) Vuong–Lo–Mendell–Rubin Likelihood Ratio Test (*VLMR*), (e) Parametric Bootstrapped Likelihood Ratio Test (*BLRT*), and (f) the practical usefulness of the solution (e.g., the number of individuals in each class, the values of estimated parameters).

2. Results

The sample correlation matrix, means and variances of the observed continuous variables are shown in Table 1. Correlations between the variables measuring school burnout range from .38 to .65 suggesting moderate stability. The overall LGM for school burnout for the whole sample was estimated first. LGM estimates different growth components, such as level (*intercept*), linear change (*linear trend*), and quadratic change (*quadratic trend*), from data consisting of at least three measurements. Besides estimating the means of these components the shape of change, LGM estimates also the variances of the level and growth components, and whether these variances are statistically significant. In the overall latent growth model, the loadings of the observed variables across Time 1 to Time 4 were fixed to 1 on the level factor and to 0, 1, 2 and 5 on the linear change factors. The residual variances were allowed to be freely estimated. If the results showed that the linear trend was not sufficient to describe the change in school burnout at the mean level, then the quadratic change component with factor loadings of 0, 1, 4 and 25 was added into the model, but because the model fit was better without the quadratic slope, it was not included.

Initial exploration of the data indicated that the linear growth models accounted for the data well. The results for the LGM ($\chi^2(5, N = 863) = 19.62, p = 0.002, CFI = 0.96, RMSEA = 0.06, SRMR = 0.04$) fitted the data well and are presented in Table 2. The results for the mean structure (Table 2) showed that when the initial level of school burnout was high, there was less increase later on. Moreover, the results for covariance structure showed statistically significant individual in both the level and linear trend of school burnout.

To examine what kinds of developmental trajectories adolescents show in terms of school burnout across the transition to an academic or vocational track, GMM analyses were applied to school burnout. These analyses aimed to explore whether distinct latent trajectory groups (groups of homogeneous subjects) could be identified on the basis of the development of school burnout across the transition to post-comprehensive education. The overall latent growth model was used as a basic model for searching for the optimal number of latent groups. Table 3 shows the fit indices and class frequencies for the compared growth mixture solutions. Comparison of the different solutions for school burnout suggested that the four-latent-group solution fitted the data best.²

The result of the final GMM for school burnout is presented in Table 4. Fig. 1 displays the estimated growth curves for the four different latent trajectories of school burnout³ and Fig. 3 displays the variances.

² The multiple group analyses with the Mplus knownclass option indicated that the results were similar for students following vocational or academic track.

³ The analyses were also performed separately using the same time points as for the students in Study 2 cohort. These results were similar to those in Study 2, suggesting high-increasing and moderate-stable groups of burnout among high school students.

Table 2

Study 1: latent growth model for the whole sample (unstandardized estimates; standard errors in parentheses).

Growth components	Whole sample ($N = 611$)
<i>Mean structure</i>	
Level (α_1)	3.50 (0.17)***
Linear slope (α_2)	0.20 (0.06)***
<i>Covariance structure</i>	
Variance of the level (ψ_{11})	0.50 (0.05)***
Variance of linear change (ψ_{22})	0.03 (0.05)***
Covariance (level, linear change; ψ_{21})	−0.33 (0.07)***
<i>Residual variances</i>	
School burnout T1 (θ_1)	0.37 (7.32)***
School burnout T2 (θ_2)	0.62 (0.03)***
School burnout T3 (θ_3)	0.48 (0.03)***
School burnout T4	0.11 (0.09)

Note 1.

*** $p < .001$.

Table 3Study 1: fit indices and class frequencies for growth mixture models for school burnout with a different number of latent trajectory groups ($N = 863$).

No. of groups	BIC	ΔBIC	AIC	Entropy	VLMR	BLRT
1	6633.97	6611.65	6600.65			
2 ($n_1 = 95\%$, $n_2 = 5\%$)	6611.73	6579.98	6564.13	0.88	0.00	0.00
3 ($n_1 = 69\%$, $n_2 = 7\%$, $n_3 = 24\%$)	6571.95	6530.66	6510.06	0.63	0.00	0.00
4 ($n_1 = 3\%$, $n_2 = 60\%$, $n_3 = 29\%$, $n_4 = 8\%$)	6571.65	6520.84	6495.49	0.66	0.00	0.00
5 ($n_1 = 3\%$, $n_2 = 1\%$, $n_3 = 30\%$, $n_4 = 7\%$, $n_5 = 59\%$)	6580.37	6520.03	6489.92	0.70	0.36	0.03

Because GMM is sensitive to outliers, additional analyses were carried out with new burnout variables the extreme values of which were excluded based on the IBM SPSS 20 outlier analyses. The results with the modified burnout variables were similar to the findings of the original analyses, suggesting a similar final latent class solution. The results (Table 4) showed that the first latent group (60%) was characterized by a low initial level, which was stable over time. The majority of the adolescents belonged to this latent group. The second latent group (8%) was characterized by a very high initial level, which decreased over time. The results for adolescents in the third group (29%) showed that the initial level of school burnout was low, but increased over time. The fourth latent group (3%) was characterized by a low initial level of school burnout which strongly increased over time. The latent groups were labeled as follows: *low-stable*, *high-decreasing*, *low-increasing*, and *low-strongly increasing school burnout trajectories*.

Next, educational expectations and then gender, track, and interaction between gender and track were added separately to the previous model as covariates. The resulting group memberships remained similar to those in the previous model with respect to the size and interpretation of the latent groups. The results showed that adolescents who had a lower level of educational expectations were more likely to belong to the high-decreasing than low-increasing or low stable groups (Table 5). Moreover, girls were over presented in the high-decreasing group compared to the low-stable group, whereas boys were over presented in the low-stable group compared to the high-decreasing group.

Finally, to examine whether school burnout trajectory membership would be related to young adults' fulfilled educational expectations later on, young adults' current educational status was added into the previous model. Again, the resulting group memberships remained similar to those in the previous model with respect to the size and interpretation of the latent groups. The results showed that young adults in the low-increasing group were more likely to have fulfilled their educational expectations than those in the high-decreasing or low-stable

trajectory groups, and that young adults in the low-stable trajectory group were more likely to have fulfilled their educational expectations than those in the high-decreasing group.

3. Study 2

The data (second cohort of the FinEdu) were collected from all the upper secondary schools in the same city as in Study 1, first when the students were in the 11th grade (January 2004) and again the following year when they were in the 12th grade (January 2005). These 17-year-old participants ($M = 17.05$, $SD = 0.27$) completed self-report questionnaires on school burnout during regular class sessions. Of the 614 students participating in the study at the first measurement point, 519 students (response rate 85%) were present for both administrations of the questionnaire.

3.1. The measures were the same as used in Study 1

The Cronbach's alpha reliabilities for the School Burnout inventory were: Time 1 = .85; Time 2 = .85. *Educational status* (Time 3, year 2008) was measured by asking about educational qualifications, current life situation and educational status. The answers were coded as follows: 1 = finished university studies, 2 = currently studying at the university, 3 = finished polytechnic, 4 = currently studying at polytechnic, 5 = currently studying for a second post-comprehensive degree, 6 = finished high-school but is not studying further, and 7 = did not finish high-school.

4. Results

The correlation between the students' burnout variables at Time 1 (Age = 17, $M = 2.77$, $Var = 0.71$) and Time 2 (Age = 18, $M = 2.74$, $Var = 0.80$) was 0.57, $p < .001$, suggesting moderate stability over time. The overall path model showed that school burnout showed

Table 4

Study 1: final latent growth model for the whole sample (unstandardized estimates; standard errors in parentheses).

Growth components	Low-stable (60%)	High-decreasing (8%)	Low-increasing (29%)	Low-strongly increasing (3%)
<i>Mean structure</i>				
Level (α_1)	2.33 (0.05)***	3.91 (0.19)***	2.42 (0.08)***	1.89 (0.24)***
Linear slope (α_2)	−0.03 (0.01)**	−0.38 (0.03)***	0.20 (0.02)***	0.53 (0.04)***
<i>Covariance structure</i>				
Variance of the level (ψ_{11})	0.36 (0.03)***	0.36 (0.03)***	0.36 (0.03)***	0.36 (0.03)***
Variance of linear change (ψ_{22})	0*	0*	0*	0*
Covariance (level, linear change; ψ_{21})	0*	0*	0*	0*
<i>Residual variances</i>				
School burnout T1 (θ_1)	0.26 (0.03)***	0.26 (0.03)***	0.26 (0.03)***	0.26 (0.03)***
School burnout T2 (θ_2)	0.72 (0.07)***	0.72 (0.07)***	0.72 (0.07)***	0.72 (0.07)***
School burnout T3 (θ_3)	0.42 (0.03)***	0.42 (0.03)***	0.42 (0.03)***	0.42 (0.03)***
School burnout T4 (θ_4)	0.07 (0.02)**	0.07 (0.02)**	0.07 (0.02)**	0.07 (0.02)**

Note.

*** $p < .001$.** $p < .01$.

* 0 = fixed to zero.

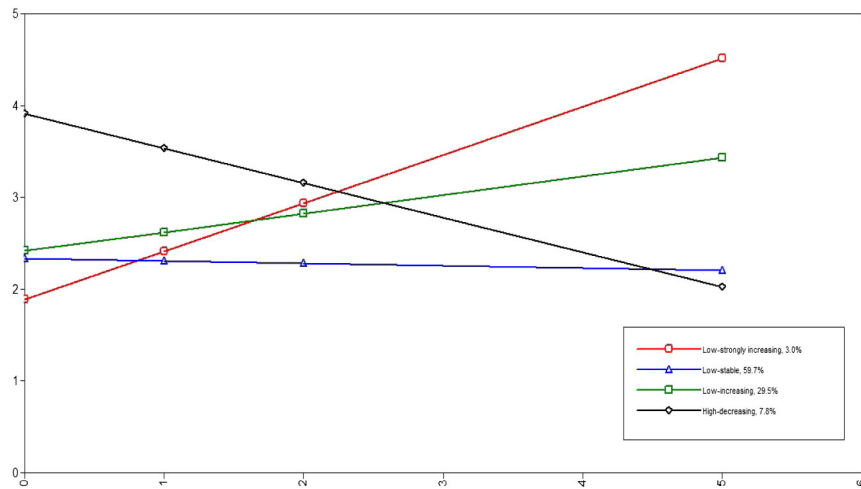


Fig. 1. Estimated growth curves for different latent trajectories of school burnout across the transition to post-comprehensive education (Study 1).

substantial stability (*standardized estimate* = 0.56, $p < .001$). The model is saturated and thus no fit indexes are given.

GMM was applied also to school burnout in Study 2. Table 6 shows the fit indices and class frequencies for the compared group solutions. Comparison of the different solutions for school burnout suggested that the two-class solution fitted the data best. Fig. 2 displays the estimated paths for the two different latent trajectories of school burnout. The results showed that most of the participants (94%) were in a moderate-slightly decreasing school burnout trajectory, while 6% were in a high and increasing school burnout trajectory. The groups were labeled *moderate-slightly decreasing trajectory* and *high increasing school burnout trajectory*. Fig. 2 displays the two individual trajectories and Fig. 4 the variances.

Next, latent change scores were generated to estimate the change in young adults' burn out between Time 1 and Time 2. The results showed that in the moderate-slightly decreasing group the slight negative change was statistically significant ($B = -0.57$, $p < .001$), whereas the change in the high increasing group was non-significant ($B = -0.15$, p ns).

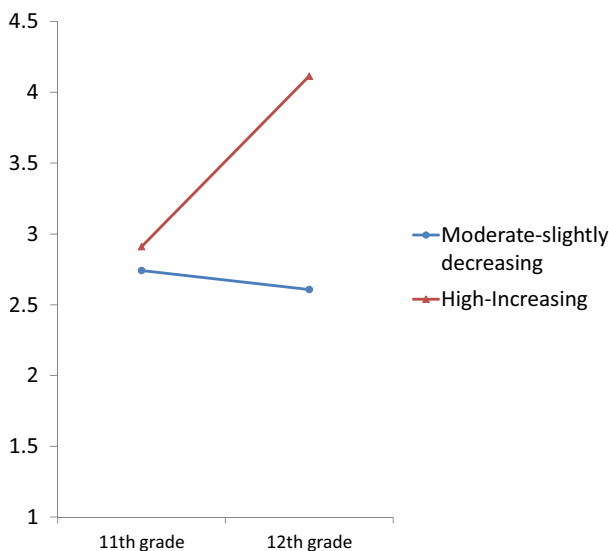


Fig. 2. Estimated growth curves for different latent trajectories of school burnout across secondary education (Study 2).

Next, educational expectations (Time 1) and gender were added separately to the previous model as covariates. The resulting group memberships remained similar to those in the previous model with respect to the size and interpretation of the latent groups. No differences were found between the latent groups. Finally, to examine whether school burnout trajectory membership would be related to their educational outcomes, young adults' educational status (Time 3) was added into the previous model. Again, the resulting group memberships remained similar to those in the previous model with respect to the size and interpretation of the latent groups. The results showed that 53% of the students in the high-increasing group were currently studying at a polytechnic (Table 7), whereas the young adults in the moderate-slightly decreasing group were more evenly distributed among the different educational institutions. However, no differences were found between the two latent trajectory groups according to the final educational status.

5. Discussion

We revealed that the development of school burnout, particularly during the educational transition period to post-comprehensive education, showed substantial heterogeneity, whereas during stable educational context, during the post-transitional period mostly stability in school burnout was found. Four trajectories were identified during the transition from comprehensive school to secondary education: low-stable, high-decreasing, low-increasing and low-strongly increasing school burnout. In turn, in upper secondary education one large quite stable trajectory, and one small stable trajectory (6%) were identified. These results supported the demands–resources model (Demerouti et al., 2001; Salmela-Aro & Upadaya, 2014). During an educational transition, the demands that students face change; accordingly, alongside stable trajectories we also identified changes (increasing and decreasing) in the trajectories of school burnout, while during the subsequent stable educational stage most were in the same stable trajectory of school burnout. The onset of secondary school entry is a significant turning point in the lives of many because of the opportunity to disengage from school and the long-term consequences of doing so. It is not an overstatement to assert that school burnout shapes life-course trajectories. School involvement predicts rates of graduation and success in college, which are important determinants of employment options.

The first aim was to examine what kinds of developmental trajectories adolescents show in terms of school burnout during the transition to post-comprehensive education. In support of the demands–resources model (Hypothesis 1b), we identified three trajectories in which school

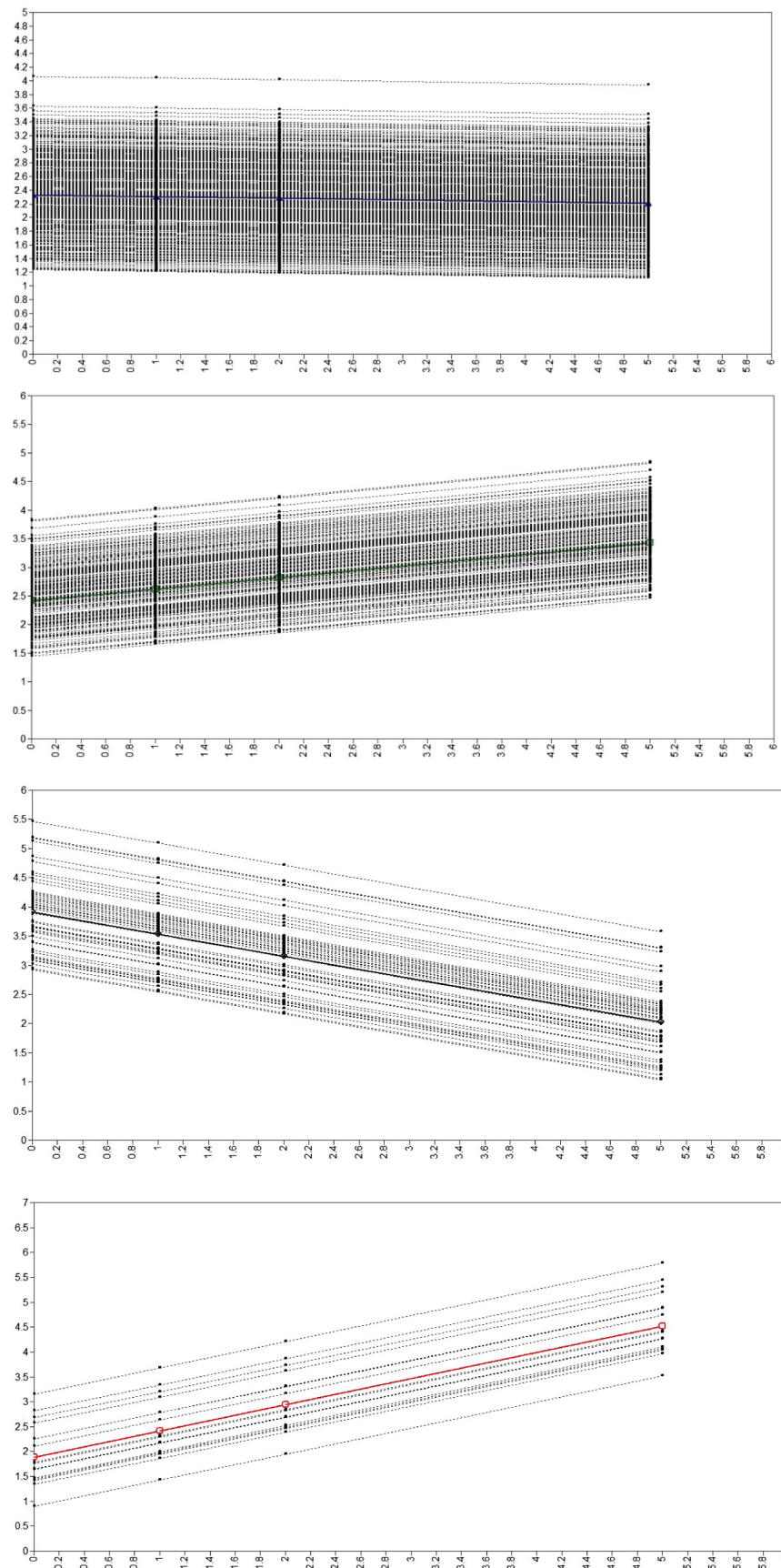


Fig. 3. Variances for the four latent trajectories of school burnout (Study 1).

Table 5

Estimated odds ratios for educational expectations (time 1), gender, and fulfilled educational expectations (time 5) of burnout latent trajectory group membership.

Comparisons	Expectations OR	Gender OR	Fulfilled expectations OR
High-decreasing vs. low-increasing	0.68**	0.22	−0.92***
High-decreasing vs. low-stable	0.64**	−1.43*	−1.42***
Low-increasing vs. low-stable	−0.05	−4.86	0.65***

Note.

* $p < .05$.

** $p < .01$.

*** $p < .001$.

burnout changed during the transition. First, a group of adolescents (28%) showed initially low but increasing level of school burnout, while another group of adolescents (8%) was characterized by a high initial but then decreasing level of school burnout. In addition, among 3% showed a low and strongly increasing school burnout trajectory. These results broaden the existing knowledge of the developmental trajectories of school burnout during the transition to post-comprehensive education by showing that both positive and negative developments in school burnout occur among different subgroups of adolescents during this period.

In accordance with Hypothesis 1a, the results showed that the vast majority of adolescents (60%) belonged to the low-stable latent trajectory group. This finding support earlier results according to which the majority of young people manage adolescence without severe problems, while some have difficulties in adapting to the transitions and changes that characterize this life phase. In the present study, school burnout increased among 31% of the participants during the transition. Previous studies among young people in the US have shown that, on average, well-being among children and adolescents typically decreases during educational transitions. Moreover, it was more typical for girls to belong to the high-decreasing group than for boys, whereas boys more typically than girls belonged to the low-stable group of burnout.

Our results suggest that, on average, the Finnish school system does relatively well in terms of the demands it imposes on the changing resources of young people during the transition to post-comprehensive education. Although the majority of the adolescents were in the latent trajectory characterized by a low level of school burnout, three other smaller latent trajectory groups were also identified in the sample. In line with our expectation, a single overall growth trajectory would have oversimplified the heterogeneity of the changes in adolescents' school burnout during the transition to post-comprehensive education.

Supporting our Hypothesis 2c, the second study during the post-transitional educational stage suggested that a small group of adolescents (6%) suffered high and increasing school burnout, while most (94%) passed through the same period with no such thoughts and feelings. This supports Hypotheses 2a and 2b in line with the demands–resources model: during the post-transitional period the trajectories mostly differs in level. Negative changes result from the fact that increasing demands in relation to students' resources lead to increasing trajectories of burnout. It is not the transition itself that is the cause of the decline in engagement; instead, it is the new demands presented by school (i.e., its climate, demands, and opportunities) in the context of adolescents' resources

(i.e., self-esteem, support, self-efficacy). Thus improving the fit of educational environment with the students' resources may be an effective way to reduce the level of disengagement. The results revealed destabilization of school burnout after the transition period.

According to the state–environment theory (Eccles & Midgley, 1989) depending on the ability of the new school environment to meet the individual needs of the students, their well-being may either improve or worsen during the transition. As most students are in the stable and low burnout trajectory it seems that in the Finnish school system there is quite a good fit between the demands and the secondary school context. However, the result might reflect students' lack of personal and social capital (Tynkkynen, Nurmi, & Salmela-Aro, 2010). Such capital would support them during the challenging developmental transition and promote their positive adaptation. Accordingly, these students may need more individualized support to help them deal with challenges.

Why are some students presenting decreasing burnout during the transition? According to demands–resources model (Salmela-Aro & Upadyaya, 2014), it might be that for this group of students they experience better fit between the demands and resources. It might be that the changing school context is less demanding and/or in the new school context they have better achieved their school-related goals and thus achieved higher self-efficacy and related resources. It might also be that the new school context offers better support and thus their burnout decreases.

Clearly, there is a need for a more integrative developmental–contextual approach to investigate the individual and contextual factors that are predictive of student burnout and to examine whether these facets of burnout are more or less important depending on the characteristics of adolescents or the ecological context in which they live (Wang, Willett, & Eccles, 2011). Due to the low stability of school burnout across educational transitions, these transitions may also be a fruitful phase in which to carry out interventions to promote adolescents' career planning and well-being, and to prevent possible negative outcomes as a result of the transition. Burnout is malleable; it is presumed to be a function of both the individual and the context. Thus it can be shaped more easily than an individual or general trait. Research on burnout can inform possible leverage points for a wide range of intervention targets aiming to improve student adjustment. Early interventions should target adolescents showing increasing school burnout in order to prevent the accumulation of problems. Researchers, educators, and policymakers should increasingly focus on burnout as a framework for tackling problems of student boredom and alienation, low achievement, and high drop-out rates, as it holds out the prospect of multiple benefits for research and practice.

Adolescents may be assumed to develop optimally and to show high levels of well-being when the requirements of their academic environments are commensurate with their resources and skills. On this reasoning, an environment that fits poorly with adolescents' resources and needs leads to negative motivational consequences for individuals, and therefore decreases their well-being (Eccles & Midgley, 1989). The results showed further that adolescents with lower level of educational expectations were more likely to belong to the high-decreasing than to the low increasing or the low stable groups. Young adults in the low-increasing group, in turn, had their educational expectations fulfilled at the end of the study more often than young adults belonging to the

Table 6

Fit indices and class frequencies for mixture models for burnout trajectories with different numbers of latent classes (Study 2).

No. of groups	BIC	ΔBIC	AIC	Entropy	VLMR	BLRT
1	1082.77	1073.25	1070.18			
2 ($n_1 = 6\%$, $n_2 = 94\%$)	1083.95	1064.90	1058.78	0.83	0.68	0.00
3 ($n_1 = 65\%$, $n_2 = 32\%$, $n_3 = 4\%$)	1083.91	1055.34	1046.16	0.49	0.12	0.00

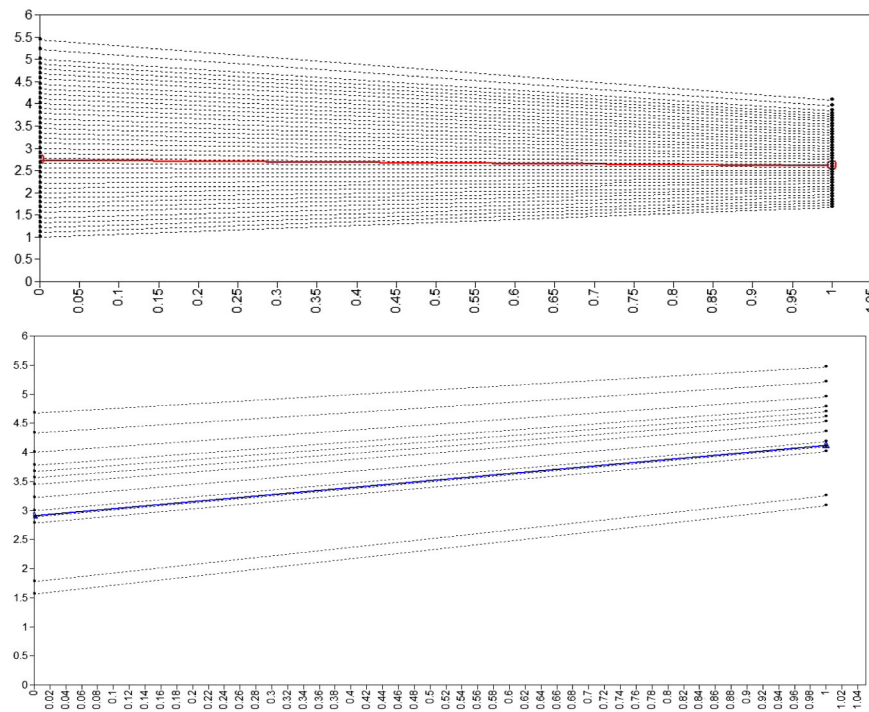


Fig. 4. Variances for the two latent trajectories of school burnout across secondary education (Study 2).

high-decreasing or low-stable trajectory groups. Moreover, the low-stable trajectory group had their educational expectations fulfilled more often than young adults belonging to the high-decreasing group.

The results revealed some gender differences. We found gender differences in Study 1 but not in Study 2. However, the two studies differ on the basis of gender. In Study 1 in the context of comprehensive school there are about an equal number of females and males, whereas in Study 2 among those in the academic track there are more females than males. Moreover, the gender difference seems to take place during comprehensive school, and thus Study 2 which takes place only in upper secondary school revealed no gender difference. Gender differences seem to emerge in comprehensive school, burnout being higher among girls than boys.

At least the following limitations should be taken into account in any effort to generalize the results. All the measures were based on self-report measures. Moreover, we had no information about other transitions, such as puberty, that might have occurred during the ages of interest. Future studies are needed to examine the role of burnout during other educational transitions. The study was carried out in Finland, and thus caution should be exercised in generalizing the results to other school contexts. However, many European countries have a

similar educational system, in which students attend comprehensive school and then go on to an academic or a vocational track. Finally, our results do not provide any causal explanations, and therefore more research is needed to identify possible reverse paths from school trajectories and school burnout.

Acknowledgement

The study has been supported by Academy of Finland (139168, 273361) and Jacobs Foundation.

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Table 7

The proportions for educational status at time 3.

	Moderate–slightly decreasing	High–increasing
Finished university	0.11***	0.04
Studying at university	0.35***	0.20
Finished polytechnic	0.21***	0.09
Studying at a polytechnic	0.18***	0.53***
Studying for a post-comprehensive degree	0.07***	0.08
Not studying further	0.07***	0.05
Not finished high-school	0.00	0.00

*** $p < .001$.

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